Sample Paper 8 — Class 4 Mathematics Olympiad (Level-2 Difficulty Step-Up)

Pattern: Same 35 questions / 40 marks / 60 minutes / no negative marking as Papers 1–7, but noticeably harder throughout — closer to SOF IMO **Level 2**.

Section 1 — Logical Reasoning (Q1–Q10, 1 mark each)

Q1. Find the odd one out: A) Triangle B) Square C) Pentagon D) Circle

Q2. Doctor is to Hospital as Teacher is to ___. A) School B) Student C) Book D) Blackboard

Q3. If each letter of a word is moved 2 steps backward in the alphabet (wrapping A → Y), FLOWER becomes DJMUCP. Using the same rule, what does MARKET become? A) KYPICR B) KYQICR C) KXPICR D) LYPICR

Q4. This series follows two patterns at once — the 1st, 3rd, 5th, 7th terms, and the 2nd, 4th, 6th, 8th terms. Find the missing (8th) term: 5, 90, 10, 84, 15, 78, 20, ___ A) 70 B) 71 C) 72 D) 73

Q5. Five students — Arjun, Bhavna, Chirag, Divya, Esha — sit in a row. Chirag sits exactly in the middle. Esha sits at the left end. Divya sits immediately to the right of Chirag. Bhavna sits between Esha and Chirag. Find the seating order, left to right. A) Esha, Bhavna, Chirag, Divya, Arjun B) Esha, Chirag, Bhavna, Divya, Arjun C) Bhavna, Esha, Chirag, Divya, Arjun D) Esha, Bhavna, Divya, Chirag, Arjun

Q6. Meera walks 12 m South, then 9 m East, then 12 m North, then 9 m West. Where is she now relative to her starting point? A) Back at the starting point B) 9 m East C) 12 m South D) 21 m away

Q7. If the 7th day of a month is a Sunday, what day of the week is the 30th of the same month? A) Monday B) Tuesday C) Wednesday D) Sunday

Q8. Find the odd one out: A) 9 B) 16 C) 24 D) 49 (Hint: three of these are a number multiplied by itself.)

Q9. Find the next number: 2, 5, 11, 23, 47, ___ A) 89 B) 92 C) 94 D) 95

Q10. In a line of students, Priya is 10th from the left and 15th from the right. If 4 students leave from the right end, what will be her new position from the right? A) 10th B) 11th C) 12th D) 13th

Section 2 — Mathematical Reasoning (Q11–Q20, 1 mark each)

Q11. 56,789 + 27,654 = ? A) 84443 B) 84343 C) 84543 D) 83443

Q12. 90,000 − 48,765 = ? A) 41235 B) 41135 C) 41335 D) 40235

Q13. 3,456 × 8 = ? A) 27648 B) 27548 C) 27748 D) 27638

Q14. 9,576 ÷ 18 = ? A) 532 B) 522 C) 542 D) 528

Q15. Find the HCF of 60 and 84. A) 6 B) 8 C) 12 D) 24

Q16. Find the LCM of 12 and 18. A) 24 B) 36 C) 48 D) 72

Q17. Which of these fractions is the greatest: 5/9, 7/12, 2/3, 3/5? A) 5/9 B) 7/12 C) 2/3 D) 3/5

Q18. 4 km 250 m + 3 km 900 m = ? A) 7 km 150 m B) 8 km 150 m C) 8 km 50 m D) 7 km 950 m

Q19. A rectangular field is 45 m long and 28 m wide. Find its area. A) 1,240 sq m B) 1,250 sq m C) 1,260 sq m D) 1,270 sq m

Q20. A regular octagon has a perimeter of 96 cm. Find the length of each side. A) 10 cm B) 11 cm C) 12 cm D) 13 cm

Section 3 — Everyday Mathematics (Q21–Q30, 1 mark each)

Q21. A school collected ₹93,500 from 5 classes equally. How much did each class contribute? A) ₹18,500 B) ₹18,600 C) ₹18,700 D) ₹18,800

Q22. A factory makes 2,520 toys in 9 hours, at a steady rate. How many toys does it make in 4 hours? A) 1,080 B) 1,100 C) 1,120 D) 1,140

Q23. A tank holds 4,200 L. It is filled by a pipe at 150 L per minute. How long will it take to fill the tank completely? A) 26 min B) 27 min C) 28 min D) 30 min

Q24. A bus leaves city A at 7:05 a.m., stops for 20 minutes on the way, and reaches city B at 12:10 p.m. How much time did the bus spend actually travelling (excluding the stop)? A) 4 h 25 min B) 4 h 35 min C) 4 h 45 min D) 4 h 55 min

Q25. 4 L 200 mL of one liquid is mixed with 2 L 800 mL of another, then poured equally into 5 bottles. How much does each bottle get? A) 1 L 200 mL B) 1 L 300 mL C) 1 L 400 mL D) 1 L 500 mL

Q26. Rohit buys 9 notebooks at ₹24 each and gets a discount of ₹21 on the total bill. How much does he pay? A) ₹190 B) ₹193 C) ₹195 D) ₹198

Q27. A cistern with 1,350 L of water is used for irrigation at the rate of 1/9 of its original capacity every day. How much water remains after 3 days? A) 850 L B) 875 L C) 900 L D) 925 L

Q28. A delivery van carries 4 boxes weighing 14 kg 200 g, 11 kg 800 g, 9 kg 300 g, and 13 kg 700 g. Find the total weight carried. A) 48 kg B) 48 kg 500 g C) 49 kg D) 49 kg 500 g

Q29. 1,224 mangoes are packed equally into boxes of 36 mangoes each. How many boxes will there be? A) 32 B) 33 C) 34 D) 36

Q30. A trader bought 60 kg of rice for ₹2,400 and sold all of it at ₹45 per kg. Find his total profit. A) ₹250 B) ₹280 C) ₹300 D) ₹320

Section 4 — Achievers Section (Q31–Q35, 2 marks each)

Q31. A number is first multiplied by 7, then 21 is subtracted from the result, giving 91. Find the original number. A) 14 B) 15 C) 16 D) 17

Q32. The sum of three numbers is 180. The second number is 3 times the first, and the third is 12 more than the second. Find the first number. A) 20 B) 22 C) 24 D) 26

Q33. Kavya had a certain amount of money. She spent 3/8 of it on books, then 1/4 of what remained on stationery, and was left with ₹900. How much money did she have at first? A) ₹1,800 B) ₹1,860 C) ₹1,920 D) ₹1,980

Q34. In a school, the number of girls is 4 more than twice the number of boys. There are 298 students in total, and 98 of them are boys. Find the number of girls. A) 194 B) 196 C) 198 D) 200

Q35. A sequence starts with 7 and 9, and each later term is the sum of the two terms before it: 7, 9, 16, 25, 41, ___. Find the next term. A) 60 B) 63 C) 66 D) 69

Answer key: `sample-paper-08-answer-key.md`
